

Comprehensive Foundation: BODMAS, Geometry & Word Problems Exam (Set 8)

Name: _____

Class: _____

Student ID: _____

Time: 35 Mins

Total: 100 Pts

Score: _____

Section 1: Order of Operations & Mental Arithmetic Rigor (4 pts each, 24 pts total)

Score: _____

Calculate step-by-step strictly following BODMAS. Do NOT skip intermediate steps:

(a) $18 + 5 \times 4 =$ _____

Multiply first!

(b) $(18 + 5) \times 4 =$ _____

Brackets first!

(c) $48 \div 6 \times 2 + 3^2 =$ _____

Powers first, then division/mult left-to-right.

(d) $50 - 2 \times (12 - 3 \times 2) =$ _____

Inside the brackets: 3×2 first.

(e) $24 \div (2 + 6) \times 5 =$ _____

Brackets first: $(2 + 6) = 8$.

(f) $100 - [40 - (15 + 5)] \div 4 =$ _____

Innermost brackets first.

Section 2: Perimeter, Area & Square Side Recovery (4 pts each, 20 pts total)

Score: _____

Pay close attention to units and formulas. Recall: Area of square $= s^2$, Perimeter $= 4s$:

(a) A square has an area of 49 cm^2 . _____

Side length = _____ cm

(b) A square has an area of 81 cm^2 . _____

What is its perimeter?

(c) A rectangle has perimeter 36 cm and width 6 cm. _____

Length = _____ cm; Area = _____ cm^2 .

(d) A square garden has perimeter 48 m. _____

What is its area in m^2 ?

(e) A wire of length 40 cm is bent into a rectangle with width 8 cm. Another identical wire is bent into a square. _____

Which shape has the greater area, and by how many cm^2 ?

Section 3: Metric Measurement & Scale Factor Conversions (3 pts each, 18 pts total) Score: _____

Convert accurately. Remember: 1D is $\times 10^1$, 2D is $\times 10^2$, 3D is $\times 10^3$:

(a) 3.5 m = _____
cm

(b) 2.4 m² = _____
cm²

(c) 0.05 m³ = _____
cm³

(d) 4,500 mL = _____
L

(e) 1.8 L = _____
cm³

(f) 150 minutes = _____
hours

Section 4: Speed, Distance & Time Precision (6 pts each, 18 pts total) Score: _____

1. [Time Conversion & Distance]

A school bus travels from School to the sports stadium at an average speed of 45 km/h. The trip takes 40 minutes.

- (a) Express 40 minutes as a fraction of an hour in simplest form: _____ hour.
(b) How many kilometers did the bus travel? Show formula and working:

2. [Speed Comparison]

Student A runs 600 meters in 2 minutes 30 seconds. Student B cycles 3.6 kilometers in 12 minutes.

- (a) Find Student A's speed in meters per second (m/s).
(b) Find Student B's speed in meters per second (m/s). Who is faster?

Working:

Section 5: Structured Problem-Solving Clinic (10 pts each, 20 pts total) Score: _____

1. [Water Tank Capacity & Volume]

A rectangular aquarium has a base measuring 80 cm long and 50 cm wide. It contains water to a depth of 30 cm.

- (a) Calculate the volume of water in the tank in cm³.
(b) Convert this volume to Liters (1 L = 1,000 cm³).
(c) If a decorative rock with volume 8,000 cm³ is submerged completely in the aquarium, by how many centimeters will the water level rise?

Structured 3-Step Working:

2. [Money & Ratio Allocation]

Student A and Student B received a total of ¥360 for helping set up the School Science Fair. Student A worked for 5 hours and Student B worked for 4 hours. They agreed to split the money in proportion to the hours worked.

- (a) How much money does each person receive per hour?
(b) How much money does Student A receive, and how much does Student B receive?
(c) Check: Does the sum of their shares equal ¥360?

Working & Reality Check:

Complete Answer Key & Pedagogical Notes

Section 1 (BODMAS):

- (a) $18 + 20 = 38$; (b) $23 \times 4 = 92$;
(c) $8 \times 2 + 9 = 16 + 9 = 25$;
(d) $50 - 2 \times (12 - 6) = 50 - 2 \times 6 = 50 - 12 = 38$;
(e) $24 \div 8 \times 5 = 3 \times 5 = 15$;
(f) $100 - [40 - 20] \div 4 = 100 - 20 \div 4 = 100 - 5 = 95$.

Section 2 (Geometry):

- (a) Side = $\sqrt{49} = 7$ cm;
(b) Side = $\sqrt{81} = 9$ cm $\implies$ Perimeter = $4 \times 9 = 36$ cm;
(c) Length + 6 = $36 \div 2 = 18 \implies$ Length = 12 cm; Area = $12 \times 6 = 72$ cm²;
(d) Side = $48 \div 4 = 12$ m $\implies$ Area = $12^2 = 144$ m²;
(e) Rect: $l = 20 - 8 = 12$ cm, Area = $12 \times 8 = 96$ cm². Square: side = 10 cm, Area = 100 cm². The square has greater area by 4 cm².

Section 3 (Units):

- (a) 350 cm; (b) $2.4 \times 10,000 = 24,000$ cm²;
(c) $0.05 \times 1,000,000 = 50,000$ cm³;
(d) 4.5 L; (e) 1,800 cm³; (f) $150 \div 60 = 2.5$ or $2\frac{1}{2}$ hours.

Section 4 & 5 (Speed & Applications):

- Sec 4: 1. (a) $\frac{40}{60} = \frac{2}{3}$ hr; (b) $45 \times \frac{2}{3} = 30$ km.
2. (a) 2 min 30 s = 150 s $\implies 600 \div 150 = 4$ m/s.
(b) $3,600 \text{ m} \div (12 \times 60 \text{ s}) = 3,600 \div 720 = 5$ m/s. Student B is faster.
Sec 5: 1. (a) $80 \times 50 \times 30 = 120,000$ cm³; (b) $120,000 \div 1000 = 120$ L; (c) Base Area = $80 \times 50 = 4,000$ cm². Rise = $8,000 \div 4,000 = 2$ cm.
2. (a) Total hours = $5 + 4 = 9$ h. Rate = $360 \div 9 = \text{¥}40/\text{h}$; (b) Student A: $5 \times 40 = \text{¥}200$; Student B: $4 \times 40 = \text{¥}160$; (c) $200 + 160 = 360$ (Verified).